Phytochemicals and cancer risk: a review of the epidemiological evidence.

A number of epidemiological studies have investigated associations between various phytochemicals and cancer risk. Phytoestrogens and carotenoids are the two most commonly studied classes of phytochemicals; phytosterols, isothiocyanates, and chlorophyll also have been investigated, although to a much lesser extent. Because there have been no systematic reviews of the literature on all phytochemicals and cancer risk to date, this article systematically reviews 96 published epidemiological studies that examined associations between phytochemicals and cancer risk. Most studies found null associations between individual phytochemicals and cancer risk at various sites. In addition, results from past studies have been largely inconsistent, and observed associations have been of relatively modest magnitude. The most consistent protective effects were observed for higher levels--dietary intake, serum, plasma, or urinary metabolites--of β-carotene and renal cell cancer, β-cryptoxanthin and lung cancer, isothiocyanates and lung cancer, isothiocyanates and gastrointestinal cancer, lignans and postmenopausal breast cancer, and flavonoids and lung cancer. Although elevated risk of certain cancers with higher levels of certain phytochemicals was observed, an insufficient pool of studies examining the same associations or inconsistent findings across studies limit the ability to conclude that any one phytochemical increases cancer risk. Additional research is needed to support previously identified associations in cases where only one study has examined a particular relationship. Importantly, continued research efforts are needed to evaluate the cumulative and interactive effects of numerous phytochemicals and phytochemical-rich foods on cancer risk.